SOLVED PROBLEM 2. The resistance of decinormal solution of a salt occupying a volume between two platinum electrodes 1.80 cm apart and 5.4 cm^2 in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.

SOLUTION

Here $l = 1.80 \text{ cm and } A = 5.4$

$\therefore$ cell constant $x = \frac{l}{A} = \frac{1.80}{5.4} = \frac{1}{3}$

observed conductance $= \frac{1}{32}$ mhos

Since the solution is $N/10$, $V = 10{,}000$ ml

Now, specific conductance $= x \times$ obs. conductance

$$\kappa = \frac{1}{3} \times \frac{1}{32} = \frac{1}{96} \text{ mhos}$$

or $\Lambda = \kappa \times V$

$$= \frac{1}{96} \times 10{,}000$$

$$= \mathbf{104.1 \text{ mhos cm}^2 \text{ equiv}^{-1}}$$

SOLVED PROBLEM 1. A solution of silver nitrate containing 12.14 g of silver in 50 ml of solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

SOLUTION

Weight of Ag in 50 ml of the solution before electrolysis $= 12.14$ g
Weight of Ag in 50 ml of the solution after electrolysis $= 11.55$ g

$$\therefore \text{ Fall in concentration of Ag} = 12.14 - 11.55 = 0.59 \text{ g} = 0.0055 \text{ g eq}$$

$$\text{Weight of Ag deposited in silver coulometer} = 1.25 \text{ g} = \frac{1.25}{108} \, g \text{ eq} = 0.0116$$

$$\text{Hence, transport number of } Ag^+ \left(t_{Ag^+}\right) = \frac{\text{Fall in conc. around anode}}{\text{No. of g eqvt deposited in silver coulometer}} = \frac{0.0055}{0.0116} = 0.474$$

and $\quad$ Transport number of NO_3^- ion $(t_{NO_3^-}) = 1 - 0.474 = \mathbf{0.526}$

SOLVED PROBLEM 2. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

SOLUTION

Wt. of copper in the anode liquid before electrolysis $= 0.79$ g
Wt. of copper in the anode liquid after electrolysis $= 0.91$ g
Increase in weight $= 0.91 - 0.79 = 0.12$ g
Increase in weight of copper cathode in the coulometer $= 0.153$ g

This means that if no copper had migrated from the anode, increase in weight would have been 0.153 g.

But actual increase $= 0.12$
Fall in concentration due to migration of Cu^{2+} ions $= 0.153 - 0.12 = 0.033$

$$\therefore \quad \text{Transport number of } Cu^{2+} \text{ ion} = \frac{0.033}{0.153} = 0.215$$

and Transport number of SO_4^{2-} ion $= (1 - 0.215) = \mathbf{0.785}$

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$= 1.5 \times 10$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction

$$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$$

is 1.9×10^3 atm^{-1}. Calculate the value of K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ or } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

$$K_c = \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}}$$

$$= 4.6 \times 10^4$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2, 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$\begin{aligned}[N_2] &= 3.00\ \text{mole}/5.00\ \text{L} = 0.600\ \text{M} \\ [H_2] &= 2.10\ \text{mole}/5.00\ \text{L} = 0.420\ \text{M} \\ [NH_3] &= 0.298\ \text{mole}/5.00\ \text{L} = 0.0596\ \text{M}\end{aligned}$$

Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c.

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C, we can write

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \mathbf{0.080}$$